K G HARISH PATEL

Computer Science Undergraduate | Computer Vision · 3D Reconstruction · Machine Learning · Efficient Deep Learning
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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate seeking a research internship in computer vision and machine learning, with hands-on work in 3D reconstruction, object detection and efficient on-device inference. At ProEmbSys Technologies, I build a real-time PCB defect-inspection system, turning open-ended production problems into measurable questions and evaluating every model change against baselines on precision, recall and mAP@0.5. Across projects in Structure-from-Motion, parameter-efficient LLM fine-tuning and graph-based retrieval, I follow a disciplined experimental workflow of controlled baselines, documented trade-offs and reproducible pipelines, backed by strong foundations in algorithms and computer architecture.

EDUCATION

B.E. in Computer Science and Engineering | *Maharaja Institute of Technology, Mysore* **Sep 2023 – Aug 2027 (Expected)**
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, Computer Architecture, DBMS (SQL), OOP

TECHNICAL SKILLS

Research Interests: Computer vision, 3D reconstruction, object detection, efficient edge deep learning, LLM evaluation
Research Methods: Experimental design, baseline comparisons, trade-off studies, precision/recall/mAP evaluation, dataset validation, error analysis
Languages: Python, C, SQL, JavaScript, TypeScript
ML & Vision: PyTorch, Scikit-learn, OpenCV, YOLOv8, MobileNet-SSD, TensorFlow Lite, Structure-from-Motion, LoRA/QLoRA, RAG, GraphRAG
Systems & Tools: Linux, Git, Docker, FastAPI, PostgreSQL, SQLite, NumPy, Pandas, NetworkX, AirSim, Unreal Engine, Jenkins CI/CD

EXPERIENCE

Project Intern, Edge AI & Machine Vision | *ProEmbSys Technologies Pvt. Ltd., Bengaluru* **Sep 2026 – Present**
- Developed a real-time Python/OpenCV vision system detecting 4 PCB defect classes (shorts, opens, missing solder, misalignment).
- Fine-tuned YOLOv8 and MobileNet-SSD through transfer learning, running per-class error analysis to trace and reduce false positives.
- Evaluated every model iteration against the prior baseline on precision, recall and mAP@0.5, accepting only non-regressing changes.
- Profiled latency, FPS, SRAM/Flash footprint and energy on constrained hardware, documenting accuracy-efficiency trade-offs.

Artificial Intelligence Intern | *Tech Mindsparc Innovations Pvt. Ltd., Remote* **Sep 2024 – Dec 2024**
- Built supervised classification models end to end in Python (Scikit-learn, Pandas, NumPy), from data preparation to evaluation.
- Engineered data-validation and feature-extraction pipelines on tabular datasets, improving accuracy over the baseline model.
- Designed a metric-based evaluation gate that rejected any change regressing model performance, keeping results reproducible.
- Delivered iterative experiments over 4 months on a remote Agile team, applying mentor code-review feedback every cycle.

PROJECTS

Drone Navigation & 3D Reconstruction | *Python, AirSim, Unreal Engine, OpenCV, NumPy* **Jun 2026**
- Developed a controller-based drone navigation simulation in AirSim and Unreal Engine to generate repeatable flight-path data.
- Automated a synchronized OpenCV data-acquisition pipeline pairing every RGB frame with its simulator camera pose.
- Reconstructed 3D scenes from the captured image sequences using Structure-from-Motion (SfM).
- Placed 2nd at the Innovotsava 2026 Drone Fair, judged on innovation and technical problem-solving.

LoRA / QLoRA LLM Fine-Tuning | *Python, PyTorch, LLMs, LoRA, QLoRA* **Jul 2026**
- Fine-tuned an open-source LLM with parameter-efficient LoRA and QLoRA adapters for a task-specific use case.
- Ran a controlled LoRA-versus-QLoRA comparison, profiling GPU memory and compute to document performance-resource trade-offs.
- Automated dataset validation checks for format, duplicates and quality before every training run.
- Benchmarked outputs against the pre-fine-tuning baseline on defined evaluation metrics to isolate the effect of each change.

GraphRAG Knowledge System | *Python, FastAPI, LLMs, NetworkX, Docker* **Aug 2026**
- Architected an LLM pipeline extracting entities and relations from unstructured documents into a NetworkX knowledge graph.
- Benchmarked graph-based answer accuracy against a chunk-based RAG baseline, verifying LLM outputs rather than trusting them.
- Exposed retrieval and question answering through a FastAPI REST service with defined JSON request/response contracts.
- Containerized the system with Docker so experiments and demos run reproducibly across environments.

LibCheck – Library Management System | *Python, FastAPI, SQLAlchemy, SQLite, JavaScript* **Dec 2025**
- Designed a 3-tier app (JavaScript UI, FastAPI REST API, SQLite) across 4 modules, documenting every endpoint in OpenAPI.
- Modeled a relational SQLAlchemy schema that automates loans, fines and book availability with consistent records.
- Secured the API with JWT authentication and role-based access control, adding audit logging for every state change.
- Shipped 7 user-facing features end to end, including smart search, analytics dashboards and email notifications.

LEADERSHIP & CERTIFICATIONS

- Co-Founder & Content Lead, Coding Kannadiga (YouTube), Jul 2025 – Present: published 90 tutorials on Python, DSA and AI (273 subscribers).
- JPMorgan Chase SWE Job Simulation (Forage, 2025); Deloitte Cyber Job Simulation (Forage, 2025); AI Aware (Intel, 2025).